SRNDNA Ultimatum: parametric offer effects

Revised scientific figure packet | 2 November 2026 | N = 47

Scope corrected: imaging figures now concern offer-amount parametric modulators and offer-modulated network coupling. Unmodulated human-computer task responses and their age effect are removed from the result narrative and figure set.

Across-partner, full-sample test: L1 COPE11 is (computer offer slope + similar-partner offer slope + dissimilar-partner offer slope)/3. The L3 all-ones coefficient gives the adjusted mean across 25 younger and 22 older participants, with centered age and the reported controls. It is already the requested across-partner, across-age activation analysis.

Offer-related activation	FLAME Z3.1	randomise t3.1	PALM Z3.1
Positive slope: larger offers	p = .00000256; 348 vox	p = .0364; 95 vox	p = .0052; 45 vox
Negative slope: smaller offers	No corrected clusters	p = .2814; 0 vox	p = .7646; 0 vox
Right-insula positive cluster	82 vox; p = .000129	0 corrected overlap	0 corrected overlap
Existing aINS / dACC spheres	0 corrected overlap	0 corrected overlap	0 corrected overlap

What this supports: positive offer-related activation converges in posterior cortex. Retained FLAME additionally supports a right-insular positive slope, confirmed anatomically with the Harvard-Oxford atlas. It is not a convergent permutation finding and does not show greater activation for unfair/smaller offers.

FLAME here uses corrected events and all-responded-trial RT regressors, not the unrepaired submitted analysis. Its group covariates/mask predate the latest PALM harmonization.

What this does not establish: no corrected negative offer-slope cluster and no corrected dACC effect are present. The predefined anterior-insula spheres have no corrected overlap, but they do not cover all anatomical insula. These are whole-brain findings, not new targeted ROI tests.

Interpretation, age effects and remaining work

Sign matters: positive activation slopes mean greater activity for larger offers; negative slopes mean greater activity for smaller offers. A contrast between partners or age groups is a difference between slopes and does not require either slope itself to be positive. Offer size changes both fairness and own monetary payoff in this design.

Activation is not PPI: offer-modulated PPI measures how network-region coupling depends on offers. The small network main effects remain in this packet because they are parametric: pooled-post DMN similar > dissimilar p = .0424; pooled-post and partner-post ECN dissimilar > similar p = .0298/.0121. They remain timing-dependent and require anatomical/influence review.

Age findings within scope: no direct age contrast in this parametric PALM batch survives Z = 3.1. Full-RT DMN younger > older has p = .2526 at Z3.1 and p = .0468 at Z2.6. Task-wide offer younger > older has p = .1017 at Z3.1 and p = .0352 at Z2.6. The unmodulated human-computer age contrast tests a different process. Age-by-behavior interactions, including the separate parietal finding, are not part of this batch.

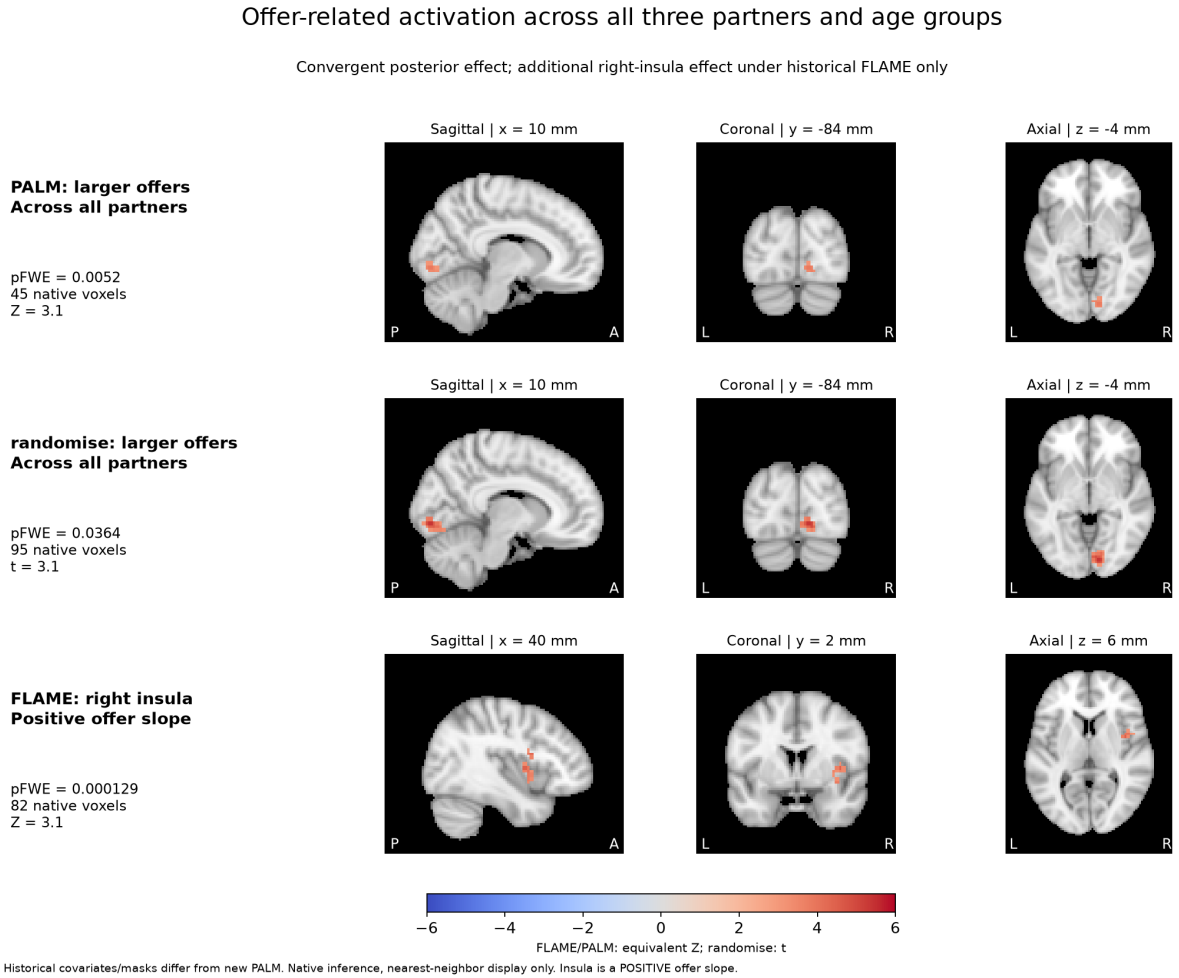
Method comparison: historical FLAME/randomise use some different motion/tSNR values and masks. PALM uses verified covariates, strict all-run coverage, age variance groups, 10,000 sign-flips and unweighted group inference. A fully matched estimator comparison remains necessary before attributing differences to software alone. FLAME is the submitted estimator; PALM is a robustness analysis.

Behavioral context retained: acceptance rises with offer size, without reliable age-by-offer-by-similarity moderation (p = .230). Ratings distinguish humans from the computer, not reliably similar from dissimilar partners. Individual norm-model parameters have not cleared recovery gates for imaging use.

Still to finish: resolve matched-design comparisons; choose the primary timing specification on design grounds; review tissue localization/influence for the parametric candidates; finalize figures and revise the manuscript/responses. No new ROI tests, small-volume correction or model changes were made to seek insula/dACC significance.

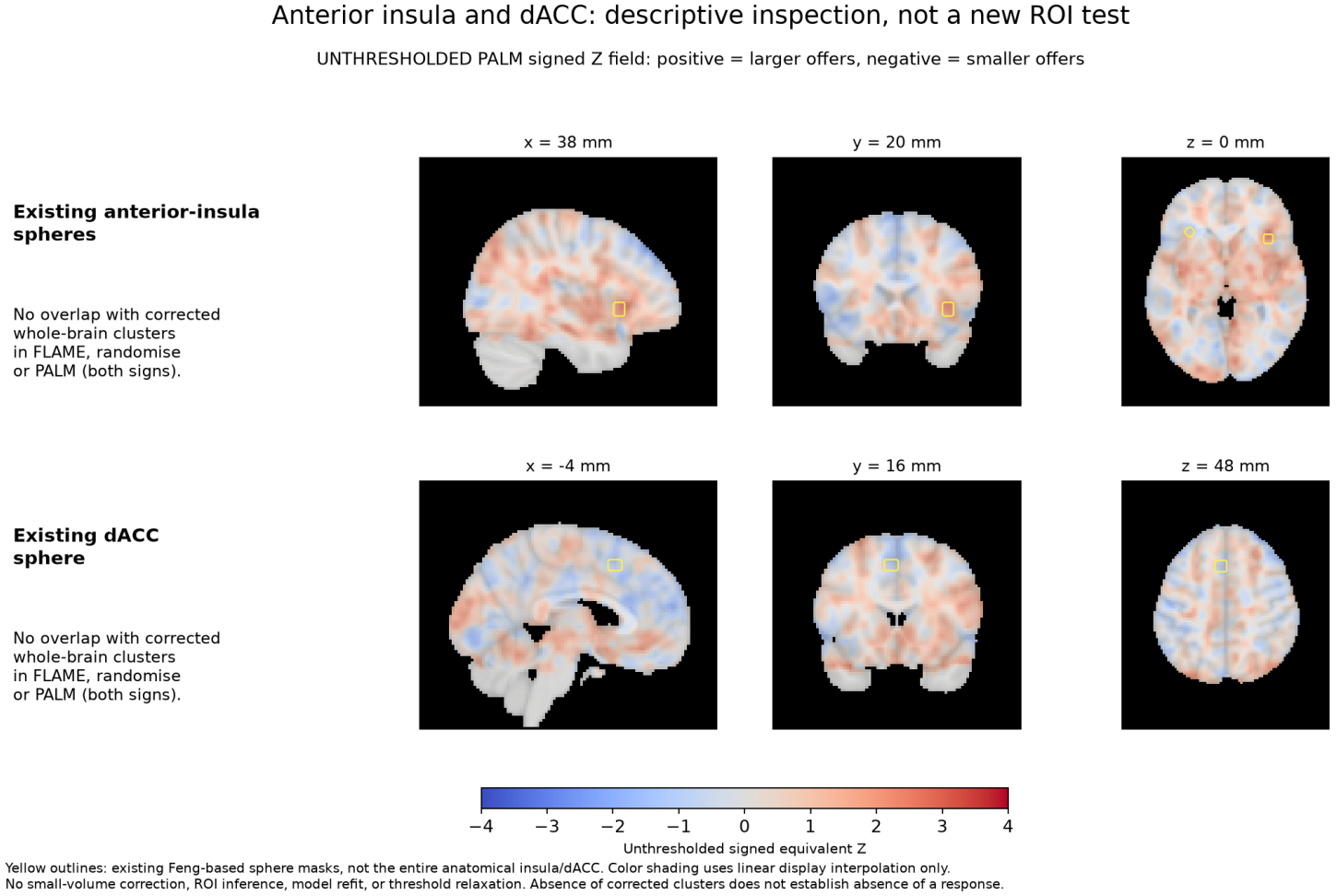
Anatomical audit: the historical right-insula cluster peaks near [40.3, 1.29, 5.22] mm and is more posterior than the existing anterior-insula sphere. Atlas percentages are descriptive template probabilities, not statistical significance or participant-specific tissue fractions. Source hashes and descriptive checks accompany the figures.

Offer activation | Collapsed across partners and age groups



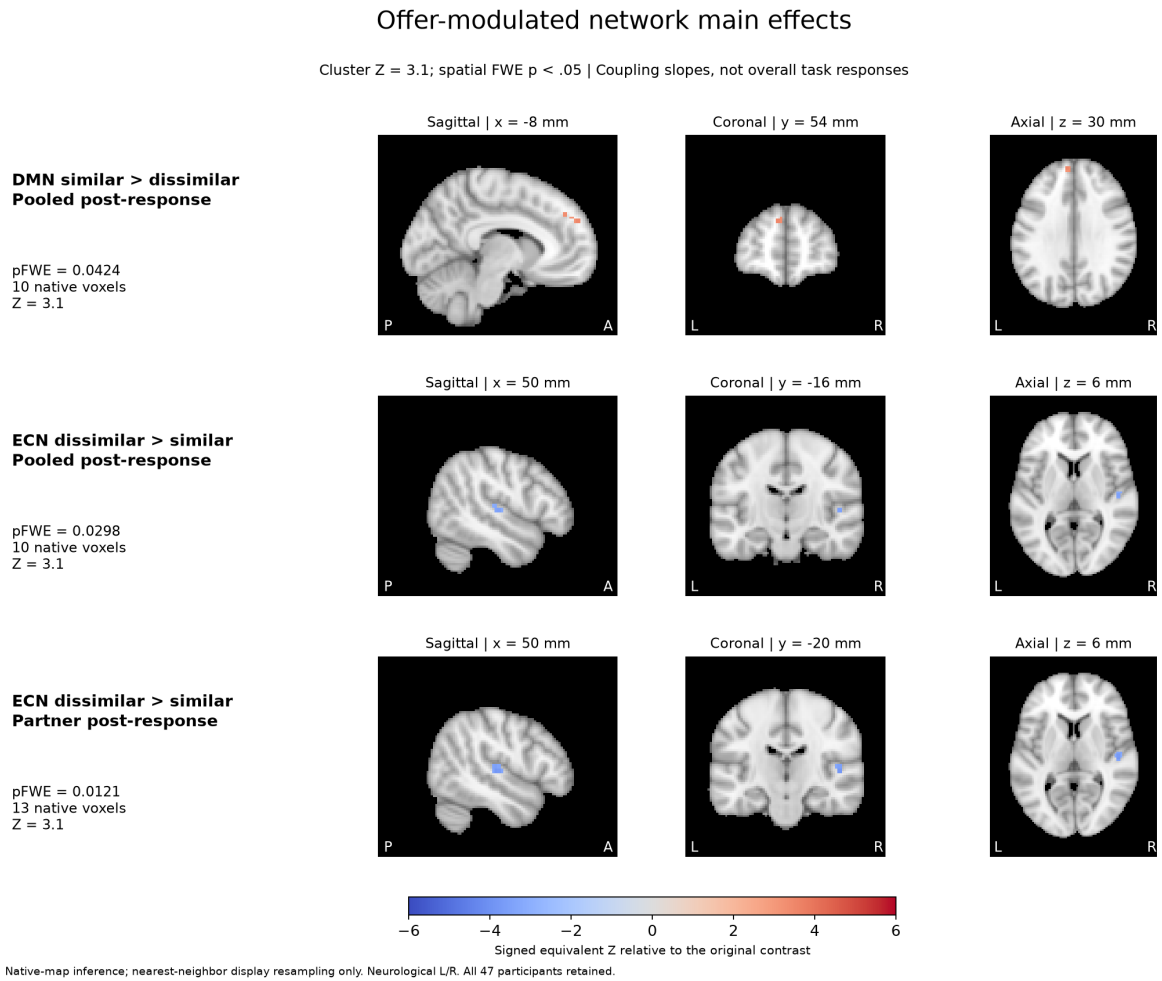
Participant COPE11 averages computer, similar-partner and dissimilar-partner offer slopes with equal one-third weights. The group intercept averages all 47 participants with age and nuisance covariates centered. PALM detects 45 posterior voxels; randomise detects 95. Historical FLAME additionally detects an 82-voxel right-insula cluster (p = .000129; peak [40.3, 1.29, 5.22]). That cluster increases with larger offers and has no corrected randomise/PALM overlap. This is not evidence of the classic smaller-offer/unfairness response. The historical FLAME map has five positive clusters, 348 voxels in total.

Insula/dACC | Existing masks and unthresholded field



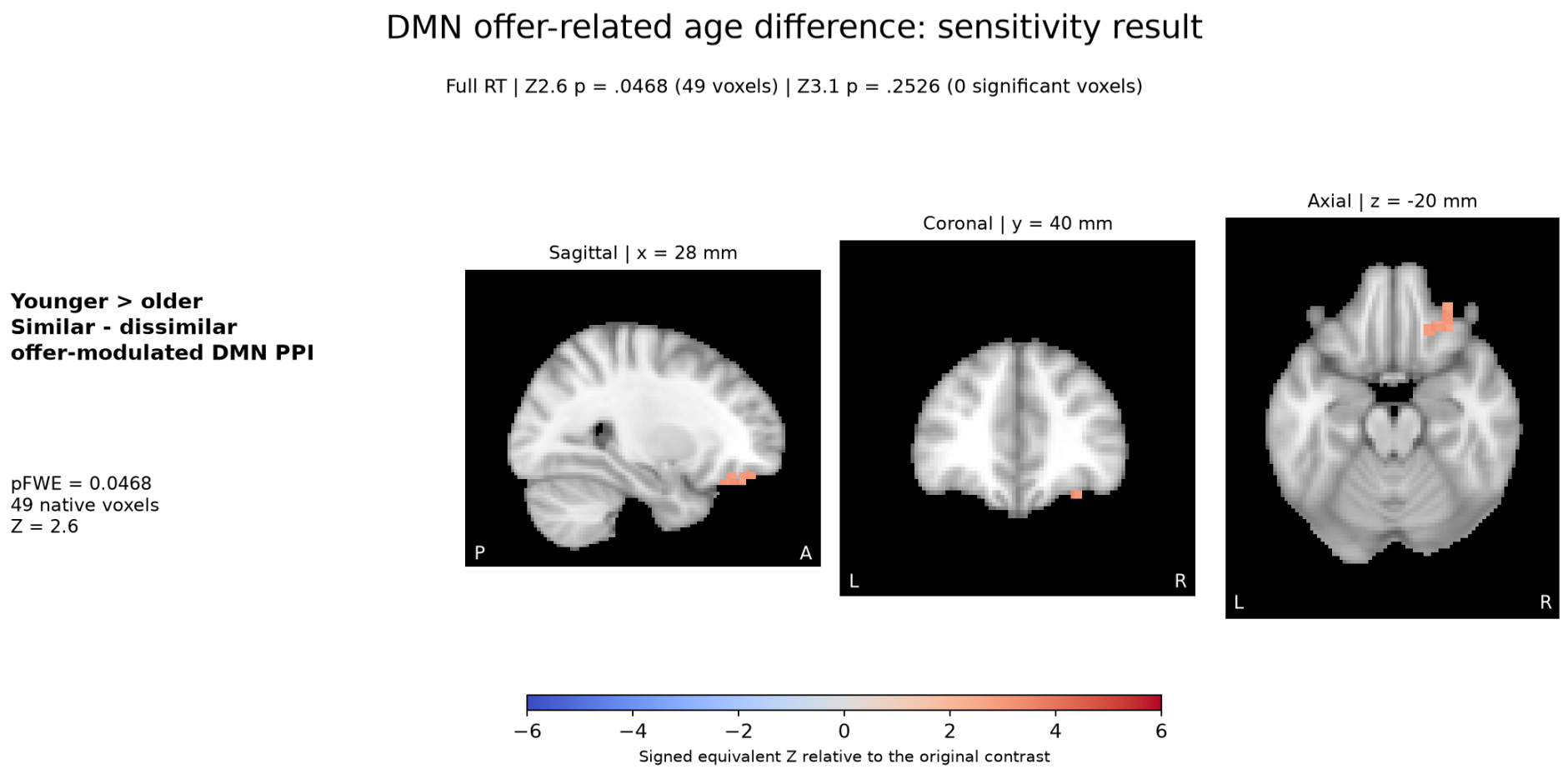
These images explain the sign and anatomical position of the existing maps; colored voxels are NOT significance claims. Neither the predefined anterior-insula spheres nor the dACC sphere contains a corrected whole-brain significant voxel in either direction. The positive historical FLAME insula cluster is more posterior than the anterior-insula spheres. No new ROI statistical test was run.

Offer-modulated network main effects



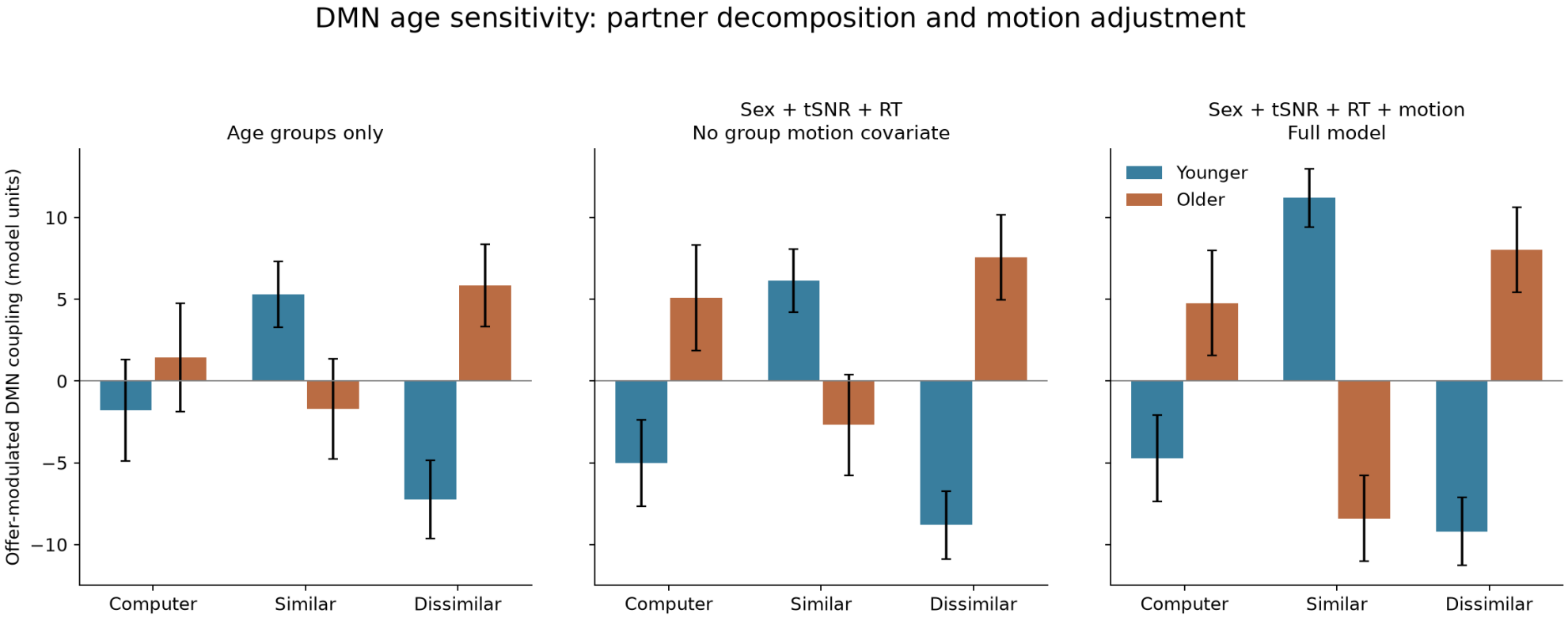
Small full-sample partner differences in offer-modulated coupling. They are not age effects and are absent at this threshold in the full-RT model. Tissue localization and influence checks remain.

DMN offer-related age difference: sensitivity result



The lower-threshold map matches the retained 49-voxel descriptive extraction region. No direct age contrast in this parametric PALM batch survives the paper-matched Z3.1 threshold.

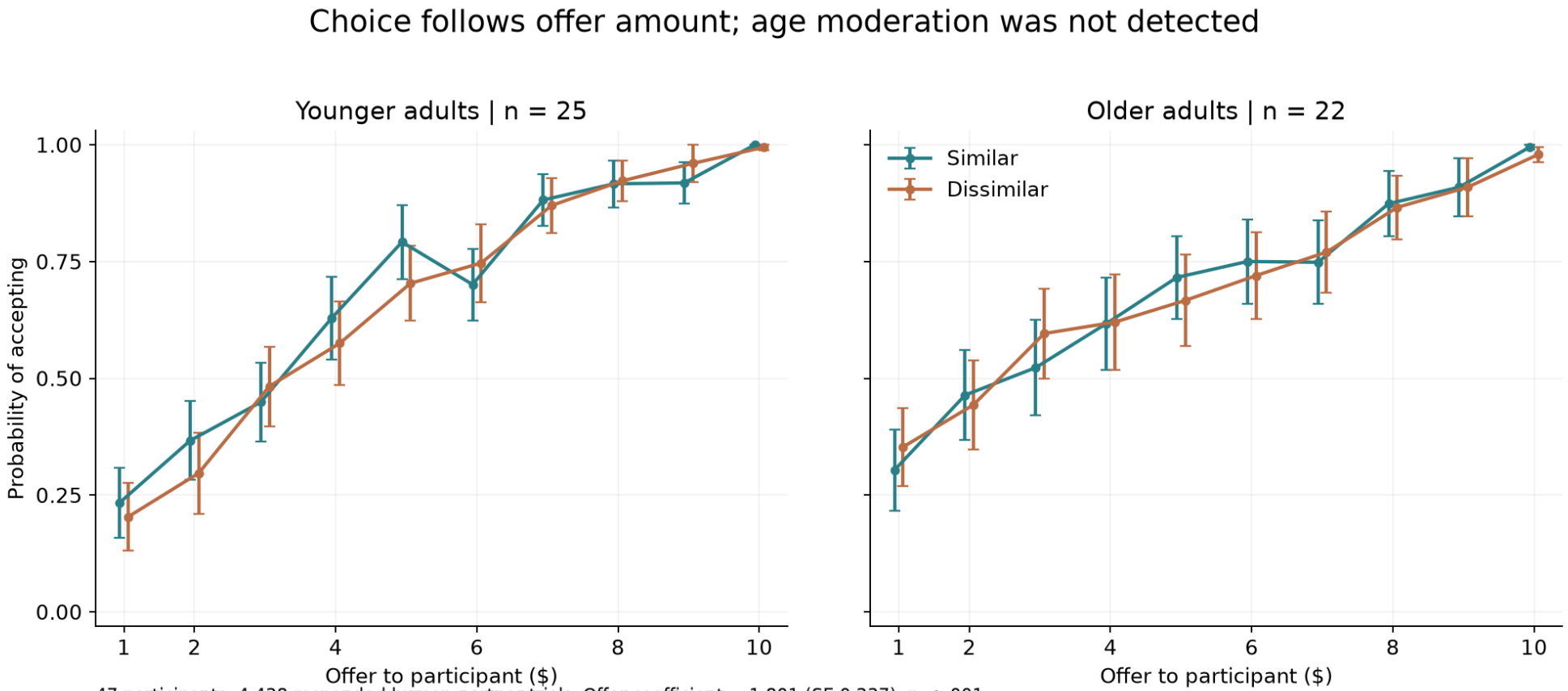
Sensitivity Figure | DMN three-partner bars



Same 49-voxel region as the new PALM Z = 2.6 age effect. Means +/- conditional participant SEM.
Descriptive selected-region summaries using common COPE7 run weights; not independent ROI tests or PALM group weights.
Younger-minus-older contrast: 20.1 age-only; 25.2 without group FD; 36.9 with all covariates.
Similar/dissimilar reverses proposer age between groups. Removing group FD does not remove L1 motion adjustment.

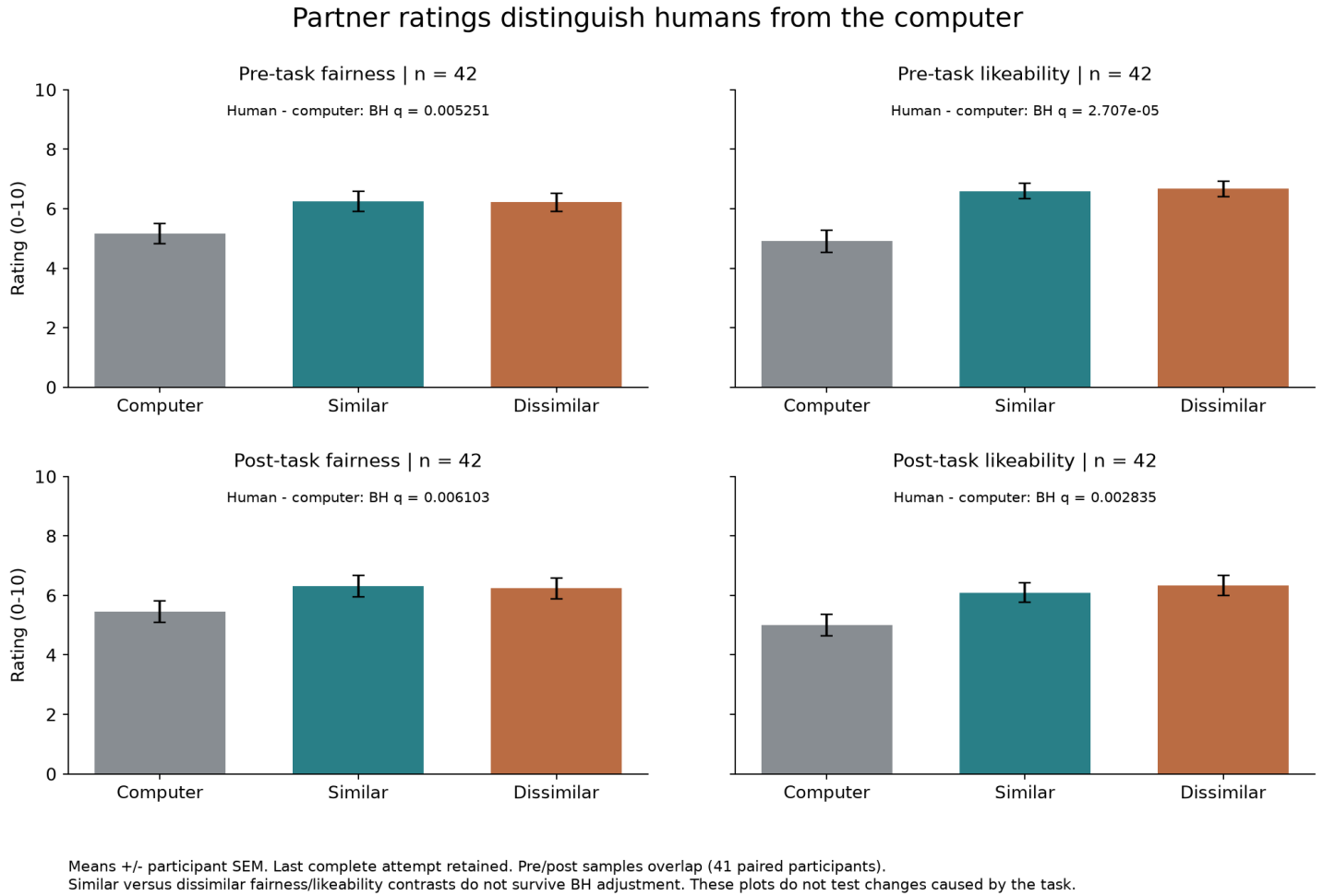
The age-gap direction is present before group motion adjustment, but its magnitude increases substantially after adjustment. The apparent crossover corresponds to positive offer-related coupling for younger proposers and negative coupling for older proposers in both participant groups. The computer condition supplies context and is not part of the similar-minus-dissimilar contrast. No new ROI significance test is performed.

Draft Figure 1 | Corrected social-choice behavior



Acceptance increased with offer size. The corrected random-intercept/random-offer-slope model converged without singularity. No reliable age or age-by-partner moderation was detected; this does not establish equivalence. Points show observed participant means and SEM, rather than a conditional logistic curve at zero random effects. Similar/dissimilar refer to proposer age relative to the participant. Offer amount also changes the participant's monetary payoff.

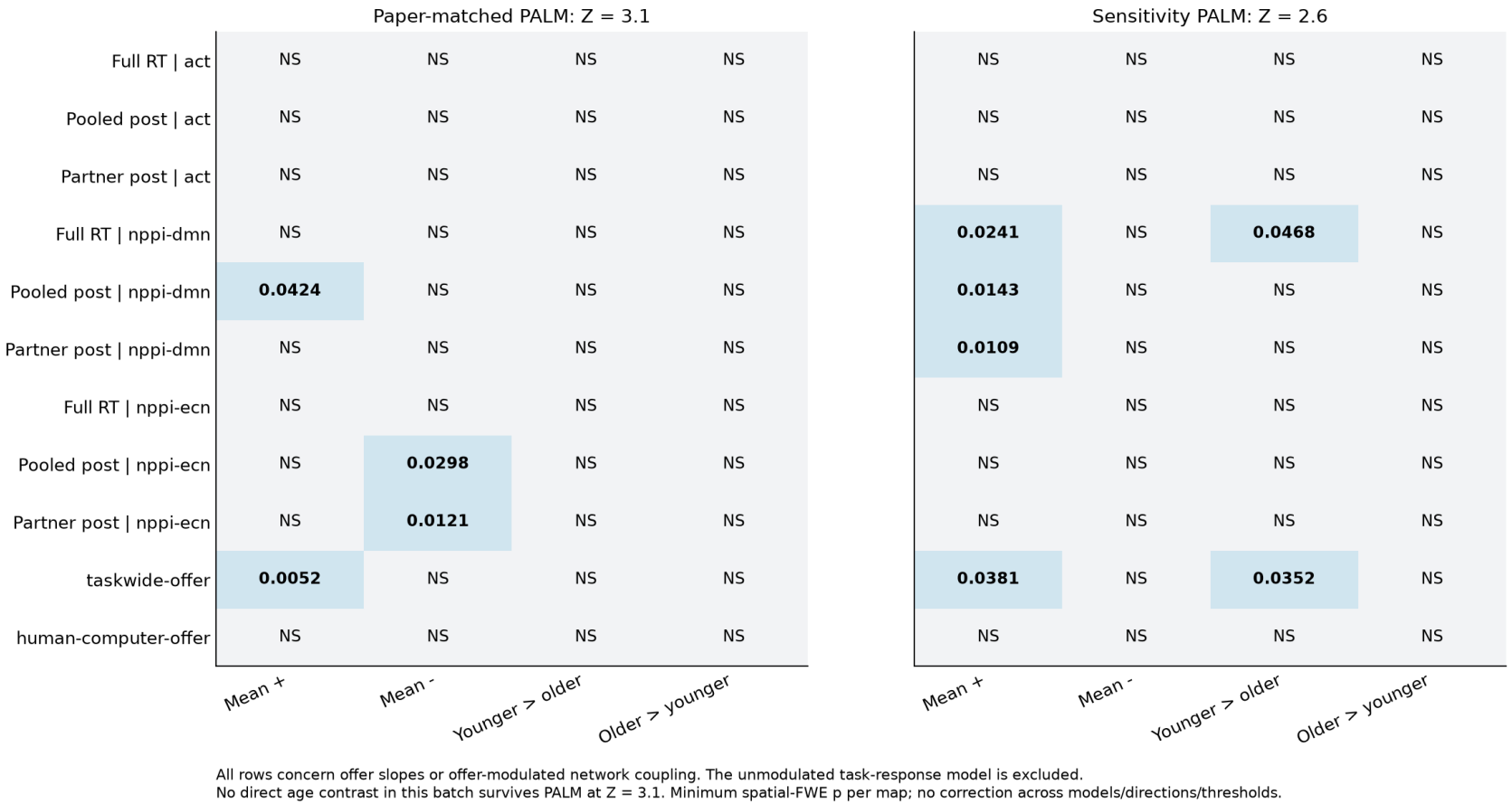
Draft Figure 2 | Partner ratings



Means +/- participant SEM. Last complete attempt retained. Pre/post samples overlap (41 paired participants).
Similar versus dissimilar fairness/likeability contrasts do not survive BH adjustment. These plots do not test changes caused by the task.

Humans were rated fairer and more likeable than the computer before and after the task. The corresponding similar-minus-dissimilar comparisons are nonsignificant after BH adjustment. These are ratings in a subsample, not evidence that partner ratings mediate an age effect.

Parametric inventory | All means and age directions



The unmodulated human-computer task-response model is removed. All 11 parametric models and both signed mean/age directions are shown. No direct age contrast in this batch survives at Z = 3.1. Lower-threshold DMN and task-wide offer age findings remain sensitivity results. Age-by-behavior interactions are separate analyses.